Internship at PES University EC Campus

Wi-Fi based microphone for conference and orientation of camera to speaker

Submitted by:

Prasun Krishna PES2UG23CS438

Under the guidance of

Prof. Bivas Bhattacharya
Professor,
PES University

2 Months

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
FACULTY OF ENGINEERING
PES UNIVERSITY
(Established under Karnataka Act No. 16 of 2013)
100ft Ring Road, Bengaluru – 560 085, Karnataka, India

PRASUN KRISHNA PESU-EC-B.Tech 2023 <pes2ug23cs438@pesu.pes.edu>

Congratulation for getting selected

1 message

CONECT center EC Campus,PESU <conect@pes.edu> Thu, May 29, 2025 at 10:36 AM
To: "CONECT center EC Campus,PESU" <conect@pes.edu>
Cc: "Dr. Sireesha B PESU-ECE" <sireeshab@pes.edu>, "Mr. BIVAS BHATTACHARYA PESSE ECE" <bivas@pes.edu>, Renuka R Karjur PESSE ECE <renukark@pes.edu>, prahalad p <prahlad@pes.edu>
Bcc: pes2ug23cs438@pesu.pes.edu

Hello

You got selected for a summer internship in CONECT.

Report to 505A on 2nd June, 2025 at 9:00am to know more details about your project.

For any doubt or clarification you can contact any of the faculties involved in CONECT.

Thanking you
CONECT
(Center Of Networking and Evolving Communication Technologies)

DECLARATION

We hereby declare that the project entitled **“Wi-Fi based microphone for conference and orientation of camera to speaker”** has been carried out at PES University by me under the guidance of Prof. Bivas Bhattacharya, Professor and submitted in partial fulfilment of the credits for the degree of **Bachelor of Technology** in **Computer Science Engineering** of **PES University, Bengaluru** during the academic semester 2 Months. The matter embodied in this report has not been submitted to any other university or institution for the award of any degree.

PES2UG23CS438	**Prasun Krishna**	**Prasun Krishna**

Acknowledgment

I would like to express my gratitude to our guide Prof. Bivas Bhattacharya, Professor PES University for their continuous guidance, assistance, and encouragement throughout the development of this project.

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I take this opportunity to thank Dr. Ajey S N R, Chairperson, Department of Electronics and Communication Engineering, PES University – Electronic City Campus, for all the knowledge and support I have received from the department.

I am deeply grateful to Prof. Jawahar Doreswamy, Chancellor, PES University, Dr. Suryaprasad J, Vice-Chancellor, PES University and Prof Nagarjuna Sadineni, Pro-Vice Chancellor, PES University for providing me various opportunities and enlightenment in every step of the way.

Finally, this internship could not have been completed without the continual support and encouragement I have received from my parents and my friends.

ABSTRACT

This report documents a two-month research internship project completed under the Department of Electronics and Communication Engineering at PES University. The project, titled *"Wi-Fi based microphone for conference and orientation of camera to speaker"*, aims to identify the active speaker in a conference setting using Wi-Fi-based distance measurement (FTM) and real-time audio streaming. The system integrates I2S microphone input, UDP-based audio transmission, and HTTP-based distance reporting to provide speaker direction estimation using two ESP32-S3 boards. This internship helped develop skills in embedded systems, wireless protocols, real-time data streaming, and system integration.

Table of Contents

1. Introduction
2. Research Environment / Faculty Lab Introduction
3. Internship Project Details
    - Project Title
    - Project Context
    - Abstract and Scope
    - Design and Technologies Used
    - Implementation Details
    - Results and Learning Outcomes
4. Conclusion
5. References
6. Appendices

1. Introduction

The internship was carried out under the guidance of the Department of Electronics and Communication Engineering, PES University, Bengaluru. It was a part of the Research Internship program and focused on embedded systems and IoT-based research. The internship spanned over two months, from June to July 2025, and was aimed at providing hands-on exposure to real-world problem-solving using microcontrollers, wireless communication, and audio processing techniques.

2. Research Environment / Faculty Lab Introduction

The research internship was conducted within PES University under the mentorship of faculty specializing in IoT and embedded systems. The lab primarily focuses on low-power wireless sensor networks, real-time data processing, and signal acquisition systems. The project contributes to ongoing research in smart conferencing solutions and indoor localization systems.

3. Internship Project Details

Project Title: *Wi-Fi based microphone for conference and orientation of camera to speaker*
Is it part of another bigger project? Yes. It contributes to a larger initiative focused on intelligent camera systems that can track the active speaker in multi-user environments such as conference rooms or classrooms.

Roles and Responsibilities

- Setup and configuration of ESP32-S3-Pico boards
- Implementation of Wi-Fi Fine Time Measurement (FTM) protocol
- Integration of INMP441 I2S microphone for capturing audio
- Streaming audio data over UDP to a laptop receiver
- Calculating distance differences from two anchors and sending them via HTTP POST
- Merging multiple embedded processes into a single system for autonomous operation
-

Project Abstract and Scope - The aim was to identify the direction of the active speaker using signal distance and voice data, all handled by ESP32-S3 boards. This system allows the camera to orient itself toward the speaker using just two boards and without any physical movement detection. The ESP32 boards run FTM ranging to calculate distances from anchors, while also capturing and transmitting audio packets over UDP.

Design Details and Technologies Used

- **Microcontroller**: ESP32-S3-Pico (Waveshare)
- **Microphone Module**: INMP441 (I2S digital mic)
- **Wi-Fi Protocols**: FTM (802.11mc), HTTP POST, UDP
- **Software Stack**: ESP-IDF v5.4.1, Visual Studio Code
- **Other Tools**: Python socket server, Wireshark (for packet debugging)

Implementation Details

- Two separate applications were first developed: one for FTM distance measurement and one for I2S microphone audio streaming
- The FTM app calculated the distance to two known MAC address anchors and sent the distance difference via HTTP POST
- The I2S app captured 16-bit audio at 16kHz and sent 30 ms chunks via UDP to a PC for logging and analysis
- Both modules were merged into one final application using FreeRTOS tasks, ensuring concurrent execution of ranging and streaming

Project Results / Learning Outcomes

- Achieved ~30 cm accuracy using Wi-Fi FTM with optimized retries
- Real-time voice data streamed reliably with minimal packet loss
- Successfully integrated HTTP, UDP, I2S, and FTM into a single firmware
- Understood multithreading using FreeRTOS in ESP-IDF
- Improved knowledge in embedded networking, I2S, and time-of-flight measurement

4. Conclusion

This research internship provided in-depth exposure to wireless technologies, embedded C programming, and real-time systems. The integration of audio processing with ranging introduced multiple concurrency challenges that were addressed using FreeRTOS task management. The successful demo confirmed the system's effectiveness in tracking speaker direction based on signal timing and audio cues. This experience has significantly strengthened skills in prototyping, debugging, and system integration.

5. Appendices

1. Espressif Systems. (2024). *ESP-IDF Programming Guide*. https://docs.espressif.com/projects/esp-idf/en/latest/esp32s3/index.html
2. INMP441 Datasheet, TDK InvenSense
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5. Waveshare ESP32-S3-Pico official pin diagram https://www.waveshare.com/w/upload/9/90/ESP32-S3-Pico-details-inter-1.jpg